

PROJECT WRITE-UP TEMPLATE

*Design and Implementation of a [Web / Mobile] Application for
[Problem Domain]*

A Guide for Final Year B.Sc. Computer Science Students
Chapters One to Five

How to use this template

This document outlines the standard chapter structure expected of a final-year Computer Science project write-up for a web or mobile application. Replace the italicized placeholder title above with your own project title (e.g., "Design and Implementation of an Online Result Checking System"). Each chapter below lists the headings and sub-headings you are expected to include, together with a short note on what each section should contain. Delete these guidance notes as you fill in your own content, and keep the numbering scheme (1.1, 1.2, 2.1, 2.1.1, etc.) consistent with your department's project guideline.

PRELIMINARY PAGES

Before Chapter One, most departments require the following pages, in this order:

- Title Page — project title, your name, matriculation number, department, and submission statement.
- Certification Page — signed by student and supervisor confirming the work is original.
- Dedication (optional, one short paragraph).
- Acknowledgements (optional, one page).
- Abstract — a single-paragraph summary (250–350 words) covering the problem, objective, method/tools used, and outcome of the project.
- Table of Contents, List of Tables, List of Figures, and List of Abbreviations (as applicable).

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Purpose: Introduce the general subject area and lead into the specific problem your app addresses. Give context on the domain (e.g., healthcare, education, banking) and how technology, particularly web/mobile applications, is applied there.

1.2 Statement of the Problem

Purpose: Clearly describe the gap, inefficiency, or challenge in the current (manual or existing digital) process that motivates the project. Be specific and evidence-based.

1.3 Aim and Objectives of the Study

State the overall aim in one sentence, then list 3–6 measurable objectives, e.g.:

- To review existing systems related to the study area.
- To design a system architecture and database for the proposed application.
- To implement the proposed [web/mobile] application using [technology stack].
- To test and validate the functionality of the developed application.

1.4 Significance of the Study

Purpose: Explain who benefits from the project (end users, institution, researchers) and why it matters.

1.5 Scope of the Study

Purpose: Define the boundaries of the project — what features/modules are included and what platform(s) it targets.

1.6 Limitations of the Study

Purpose: State constraints such as time, data availability, internet/device access, or features intentionally left out.

1.7 Definition of Terms

Purpose: Provide short definitions for technical terms and acronyms used throughout the report (e.g., API, UI/UX, CRUD, MVC).

1.8 Organization of the Report

Purpose: Briefly summarize what each of the remaining chapters (2–5) covers, in one or two sentences each.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Purpose: *Introduce the purpose of the literature review and how it is structured.*

2.2 Conceptual Framework

2.2.1 Overview of Web/Mobile Applications

Purpose: *Explain relevant concepts: client-server architecture, responsive/native/hybrid design, APIs, databases, etc.*

2.2.2 Review of Relevant Technologies and Tools

Purpose: *Discuss the languages, frameworks, and platforms relevant to your project (e.g., React, Flutter, Node.js, Firebase, MySQL) and why such tools are commonly used.*

2.3 Review of Related/Existing Systems

Purpose: *Summarize at least 3–5 similar existing systems or academic works, describing what each does and the technology/approach used.*

2.4 Comparative Analysis of Existing Systems

Purpose: *Compare the reviewed systems (a table works well here) against criteria such as cost, features, platform, and usability.*

2.5 Summary and Gaps in Literature

Purpose: *Identify the shortcomings of existing systems that your proposed application will address, justifying the need for your project.*

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

Purpose: Briefly state the purpose of this chapter.

3.2 Methodology

Purpose: Describe and justify the software development methodology adopted (e.g., Agile, Waterfall, RAD, Object-Oriented Analysis and Design).

3.3 Analysis of the Existing System

Purpose: Describe how the process currently works (manual or the system being replaced) and its workflow.

3.4 Problems of the Existing System

Purpose: List the specific weaknesses identified in Section 3.3 that justify your proposed system.

3.5 Analysis of the Proposed System

3.5.1 Functional Requirements

Purpose: List what the system should do, e.g., user registration, login, search, notifications, payment processing.

3.5.2 Non-Functional Requirements

Purpose: Describe quality attributes: performance, security, scalability, usability, availability.

3.6 System Design

3.6.1 System Architecture

Purpose: Present and describe the high-level architecture diagram (e.g., three-tier, client-server, MVC).

3.6.2 Use Case Diagram

Purpose: Show actors and their interactions with the system.

3.6.3 Data Flow Diagram / Sequence Diagram

Purpose: Illustrate how data moves through the system for key processes.

3.6.4 Database Design

Purpose: Present the Entity-Relationship Diagram (ERD) and describe database tables, fields, data types, and relationships.

3.6.5 Interface (UI/UX) Design

Purpose: Include wireframes or mockups of key screens (login, dashboard, main features) with a short description of each.

3.7 Tools, Languages and Platform Requirements

Purpose: List hardware and software requirements: programming languages, frameworks, database, IDE, minimum device/browser requirements.

CHAPTER 4

SYSTEM IMPLEMENTATION AND TESTING

4.1 Introduction

Purpose: State the purpose of this chapter: to describe how the design in Chapter Three was implemented and tested.

4.2 Implementation Environment

Purpose: Describe the hardware and software environment used to build and run the application (OS, hosting, device specs, backend/frontend stack).

4.3 Choice of Programming Language and Tools

Purpose: Justify the specific languages, frameworks, and libraries chosen (e.g., why Flutter over React Native, why MongoDB over MySQL).

4.4 Description of Modules

Purpose: Describe each major module/feature of the completed application (e.g., Authentication Module, Dashboard Module, Payment Module) and how it works.

4.5 System Testing

4.5.1 Unit Testing

Purpose: Describe testing of individual components/functions.

4.5.2 Integration Testing

Purpose: Describe testing of combined modules working together.

4.5.3 System / User Acceptance Testing

Purpose: Describe end-to-end testing and, where applicable, feedback from real or sample users.

4.6 Test Cases and Results

Purpose: Present a table of test cases: input, expected output, actual output, and pass/fail status.

4.7 System Screenshots and User Guide

Purpose: Include annotated screenshots of the working application walking through key features.

4.8 Discussion of Findings

Purpose: Reflect on how well the system met its objectives, any challenges encountered, and how they were resolved.

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of the Study

Purpose: Concisely restate the problem, objectives, and what was achieved across Chapters One to Four.

5.2 Conclusion

Purpose: State the overall conclusion drawn from the project — whether the objectives in 1.3 were met.

5.3 Contribution to Knowledge

Purpose: State what is novel or valuable about your work, however modest (e.g., a specific feature, dataset, or approach).

5.4 Recommendations

Purpose: Recommend how the system could be adopted, scaled, or improved by the institution/organization.

5.5 Suggestions for Further Studies

Purpose: Suggest features or research directions left for future work (e.g., adding AI features, scaling to more platforms).

REFERENCES

List all sources cited in the report using the referencing style specified by your department (e.g., APA).

APPENDICES

Include supplementary material such as source code listings, questionnaires/interview guides (if used), and any additional diagrams or tables too large for the main text.